

Accompanying summary for the dataset review:

- **spodumene concentrate production | AU**
- **lithium brine concentration | CN**
- **lithium carbonate production, battery grade, from concentrated brine | CN**
- **lithium brine concentration | US**
- **lithium carbonate production, battery grade, from concentrated brine | US**
- **lithium brine concentration, Fenix | AR**
- **lithium carbonate production, battery grade, from concentrated brine, Fenix | AR**
- **lithium brine concentration, Salar de Olaroz | AR**
- **lithium carbonate production, battery grade, from concentrated brine, Salar de Olaroz | AR**

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1 Introduction

This document accompanies the external review of the following datasets:

- spodumene concentrate production | AU
- lithium brine concentration | CN
- lithium carbonate production, battery grade, from concentrated brine | CN
- lithium brine concentration | US
- lithium carbonate production, battery grade, from concentrated brine | US
- lithium brine concentration, Fenix | AR
- lithium carbonate production, battery grade, from concentrated brine, Fenix | AR
- lithium brine concentration, Salar de Olaroz | AR
- lithium carbonate production, battery grade, from concentrated brine, Salar de Olaroz | AR

Additionally, it provides an overview of the current ecoinvent framework for lithium in order to give historical context, and presents the new framework and the regions that will be covered.

The document focuses on the datasets submitted to ecoinvent and supports the reviewer in understanding dataset scope, methodological choices, and consistency with the framework.

Datasets that are part of the BRGM submission will be documented in a separate report that details modelling decisions, assumptions, data sources, and any related methodological choices.

The report provides:

- Context and system boundaries for the datasets
- Key modelling decisions and deviations from original sources
- Mass balance information
- Climate change scores and comparison with literature

2 Background

2.1 Current Lithium production dataset (upstream)

The goal of this data submission is to improve the lithium supply chain in ecoinvent. The current structure of the lithium supply chain is shown below. Lithium hydroxide is then produced from lithium carbonate GLO.



Figure 1: ecoinvent v3.12 lithium framework

Here the list of the datasets currently available in ecoinvent, with link to ecoQuery:

- [Lithium brine inspissation | GLO](#)
- [Market for lithium brine, 6.7% Li | GLO](#)
- [Lithium carbonate production, from concentrated brine | GLO](#)
- [Spodumene production | AU](#)
- [Spodumene production | RoW](#) (copy of AU)
- [Market for spodumene | GLO](#)
- [Lithium carbonate production, from spodumene | CN](#)
- [Lithium carbonate production, from spodumene | RoW](#) (copy of China)
- [Market for lithium carbonate | GLO](#)
- [Lithium hydroxide production | GLO](#)

3 New framework

To address the gaps and improve geographical specificity in ecoinvent's lithium supply chain, we have developed a comprehensive framework that captures the diversity of lithium production pathways globally, with recent data from literature. This framework distinguishes between two primary lithium extraction routes, hard-rock mining (spodumene) and brine extraction, and models region-specific processing chains that reflect current production methods and geographical representation.

The framework captures the complete supply chain for lithium carbonate production from both primary extraction routes. For brine-based sources, this includes evaporation pond operations to concentrate lithium-bearing brines, followed by chemical conversion to produce battery-grade lithium carbonate (Li_2CO_3). For hard-rock sources, the chain covers mining operations and beneficiation to produce spodumene concentrate in Australia, and subsequent chemical conversion in China to produce battery-grade lithium carbonate from the imported Australian spodumene.

Additionally, lithium hydroxide monohydrate ($\text{LiOH}\cdot\text{H}_2\text{O}$) production is also covered in China and Chile.

The framework thus captures the complete upstream-to-midstream supply chain, enabling comprehensive life cycle assessments from raw material extraction through to the two main lithium chemical products used in battery manufacturing.

This structured approach enables more accurate life cycle assessments by:

- Providing geographically differentiated datasets for key production regions (Australia, China, United States, and Argentina)
- Distinguishing between different salar operations where multiple sites exist within the same country
- Ensuring methodological consistency across all datasets

Figure 2 and Table 1 below show the complete structure of activities and reference products. Please note that the new framework will **fully replace all the existing datasets in ecoinvent v3.12 and prior**. Only the dataset '[lithium hydroxide production](#)' will stay in the next release, as this dataset represents the anhydrous technical grade of lithium hydroxide, typically used in the production of products such as: adhesives and sealants, lubricants and greases, fillers, putties, plasters, modelling clay, hydraulic fluids and metal working fluids.

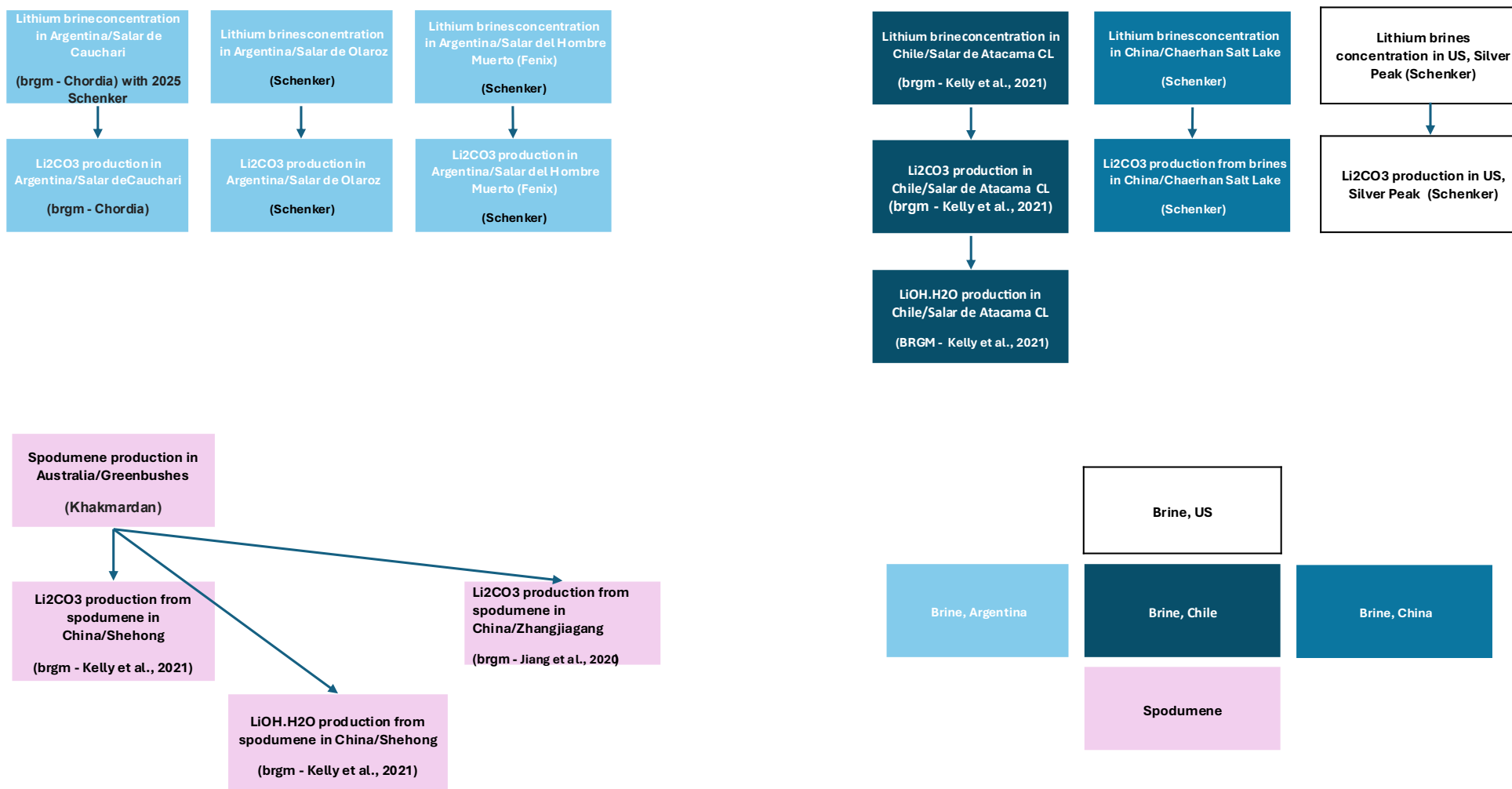


Figure 2: New lithium framework in ecoinvent

In the table below, the name of the activities and their Reference Products are reported, to ensure consistency. The name of the site is explicit in the dataset only when multiple sites exist in the same geography.

Table 1: summary of new datasets and relative reference products

Activity name	Geo	Reference Product
lithium brine concentration, Salar de Cauchari	AR	concentrated lithium brine, 6%Li
lithium carbonate production, battery grade, from concentrated brine, Salar de Cauchari	AR	lithium carbonate, battery grade
lithium brine concentration	CL	lithium carbonate, battery grade
lithium carbonate production, battery grade, from concentrated brine	CL	concentrated lithium brine, 4.3%Li
lithium hydroxide monohydrate production, from lithium carbonate	CL	lithium hydroxide monohydrate
lithium carbonate production, battery grade, from spodumene concentrate, Shenhong	CN	lithium carbonate, battery grade
lithium carbonate production, battery grade, from spodumene concentrate, Zhangjiagang	CN	lithium carbonate, battery grade
lithium hydroxide monohydrate production, from spodumene concentrate	CN	lithium hydroxide monohydrate
spodumene concentrate production	AU	spodumene concentrate
lithium brine concentration	CN	concentrated lithium brine, 0.022%Li
lithium brine concentration, Fenix	AR	concentrated lithium brine, 0.07%Li
lithium brine concentration, Salar de Olaroz	AR	concentrated lithium brine, 1.2%Li
lithium brine concentration	US	concentrated lithium brine, 0.6%Li
lithium carbonate production, battery grade, from concentrated brine	CN	lithium carbonate, battery grade
lithium carbonate production, battery grade, from concentrated brine, Salar de Olaroz	AR	lithium carbonate, battery grade
lithium carbonate production, battery grade, from concentrated brine	US	lithium carbonate, battery grade
lithium carbonate production, battery grade, from concentrated brine, Fenix	AR	lithium carbonate, battery grade

4 Allocation of submissions

The datasets included in this framework are being developed through a collaborative effort between ecoinvent and BRGM (Bureau de Recherches Géologiques et Minières). Of the 17 total datasets, BRGM will submit 8 datasets while the remaining 9 datasets will be submitted by ecoinvent.

Table 2 displays the current plan for the allocation of submissions.

Dataset	Submission done by
Cauchari Brines and carbonate AR	brgm (2 ds)
Olaroz Brines and carbonate AR	ecoinvent (2ds)
Fenix brine and carbonate AR	ecoinvent (2ds)
Atacama brines, carbonate, and hydroxide CL	brgm (3 ds)
Chaerhan brines and carbonate CN	ecoinvent (2ds)
Silver Peak US	ecoinvent (2ds)
Greenbushes spodumene AU	ecoinvent (1ds)
Shehong carbonate and hydroxide CN	brgm (2 ds)
Zhangjiang carbonate CN	brgm (1 ds)

5 ecoinvent submission

5.1 Spodumene production

The primary reference for this dataset is:

Khakmardan, S., Crawford, R. H., Giurco, D., & Li, W. (2025). Constructing a life cycle inventory of Spodumene concentrate production: Greenbushes case, Western Australia. *Journal of Cleaner Production*, 496, 145123. <https://doi.org/10.1016/j.jclepro.2025.145123>

This dataset represents the cradle-to-gate life cycle inventory for spodumene concentrate (SC6, ~6% Li₂O) produced at the Greenbushes operation in Western Australia for the year 2022.

System Boundaries and Scope

The dataset covers the complete production chain from ore extraction to the processing plant gate, with spodumene concentrate as the reference product. The system boundaries include:

- **Mining operations:** Open-pit mining with conventional drilling and blasting, overburden removal and waste handling, ore loading and on-site haulage, and all auxiliary mine operations including mine infrastructure and land use
- **Beneficiation and processing:** Crushing and screening, dense media separation using ferrosilicon, grinding and classification, flotation using fatty acid collector and glycol-based frother with pH modifiers, filtration and dewatering, and tailings management.

The functional unit is 1 kg of dewatered spodumene concentrate containing approximately 6% Li₂O. The dataset excludes downstream transport and chemical conversion processes.

Modelling choices in ecoinvent:

1. The original paper excludes the tantalum byproduct from LCIA due to its negligible mass ratio relative to SC6. In ecoinvent, economic allocation is applied for the tantalum concentrate co-product (30% Ta₂O₅); however, given the minimal mass contribution, the allocation factor remains negligible and results are comparable.
2. Compared to the paper, three specific exchanges were approximated due to the unavailability of detailed equipment-specific data in the ecoinvent database and unavailability of enough data to recreate their inventories:
 - “blasthole drilling by D65”, proxied with an addition of diesel for blasthole drilling (111MJ*0.000325 m = 0.036075MJ)
 - “excavation by CAT 6015 excavator” proxied with “excavation, hydraulic digger”
 - “haul by CAT 777 mining truck”, proxied with “transport, freight, lorry, >32 metric ton, diesel, EUR 4”

These three exchanges collectively would contribute approximately 6% (0.02 kg CO₂eq/kg spodumene concentrate) to the global warming potential. It is recommended to improve the representation of these exchanges in future database updates and this is flagged in the IncludedActivityEnd field of the dataset.

Exchange flow for road use:

“mining road” and “mining road maintenance” were created by the authors based on ecoinvent “road” and “road maintenance”. We have introduced them with the original ecoinvent names.

Mass balance

The mass balance for this dataset can be found in the Excel unit process balance (spodumene, undef).xlsx.

In the Excel document, it is possible to observe for each property whether there is a balance between the input and output, and which exchanges mostly contribute to it.

Ideally, it is advisable to allow a maximum of 1% difference between input and output for each property, with 5% as a worst-case scenario. Therefore, it is required to carefully review the properties where this limit is trespassed. Please note that wet mass = dry mass + water in wet mass

Mass balance key highlights:

- Wet mass shows an imbalance of 0.43%, which we consider acceptable
- Dry mass and water content in wet mass show an imbalance of 7.23% and 35.37%, respectively. **These imbalances were investigated:** the imbalance is driven by the default ecoinvent properties automatically assigned to the exchange “non sulfidic tailing, off-site” (30% water). The paper does not report a specific tailings composition/water content. If the tailings were assumed to be dry, the balances would close.
- **Lithium content in spodumene from ground:** Lithium is imbalanced because the Li content is based on default ecoinvent properties for “spodumene from ground”. **Action point:** We suggest balancing the lithium output via the properties of ‘spodumene from ground’ to maintain elemental closure.
- **Other elements (Al, Si, O):** we observe an imbalance because these elements are present in ecoinvent’s default “spodumene from ground” properties but not reported for the concentrate output. **Action point:** we propose either updating output ‘spodumene concentrate’ properties or adding a note in the metadata.

Scores

This section presents an estimation of the GWP100 score of the dataset. Please keep in mind that these are indicative estimations based on v3.12 framework and that the final scores will be affected by next iterations in this project and updates in other areas of the database.

Score in Khakmardan: 0.323 kg Co2 - Eq

Score in ecoinvent: 0.320 kg CO2 - Eq

Exchange name Khakmardan	Percentage contribution Khakmardan	Exchange name ei	Percentage contribution ei
Electricity, medium voltage	39.22%	electricity, medium voltage	34.38%
Grinding media (Chromium Steel)	20.53%	steel, chromium steel 18/8	24.12%
Diesel	13.65%	diesel, burned in building machine	16.42%
Blasting with explosives	6.58%	blasting	6.54%
Grinding media manufacturing (Forging)	4.68%	forging, steel	5.83%
Oleic acid	4.33%	fatty acid	2.66%
Transport by mining truck	2.91%	transport, freight, lorry, >32 metric ton, diesel, EURO 4	2.34%
Building (multi-storey)	2.27%	building, multi-storey	2.61%
Excavation by mining excavator	1.71%	excavation, hydraulic digger	0.88%
Others	4.12%	Others	4.22%
TOTAL	100%		100%

5.2 Lithium brine concentration and lithium carbonate, battery grade production

This section presents the lithium brine concentration and lithium carbonate production datasets for multiple salars. While site-specific details and considerations are discussed individually for each salar, several methodological choices and data treatment approaches are common across all brine-based datasets. These shared elements are described below before presenting the individual salars' datasets.

Data Source

The lithium brine concentration and lithium carbonate production datasets presented here are based on the modeling work of Schenker & Pfister (2025), which represents an updated and expanded version of earlier 2022 framework:

- **Schenker, V., & Pfister, S. (2025).** Current and Future Impacts of Lithium Carbonate from Brines: A Global Regionalized Life Cycle Assessment Model. *Environmental Science & Technology*, 59(13), 6543-6555.

This 2025 publication builds upon and updates the methodology presented in:

- Schenker, V., Oberschelp, C., & Pfister, S. (2022). Regionalized life cycle assessment of present and future lithium production for Li-ion batteries. *Resources, Conservation and Recycling*, 187, 106611.

Scaling Methodology: Evaporation Ponds to Concentrated Brine

Schenker & Pfister (2025) model the evaporation pond process as producing one functional unit per "unit of evaporation pond" or "unit of DLE evaporation ponds", according to the site operation mode. To integrate these inventories into ecoinvent framework, we needed to scale the evaporation pond inventories to the reference product "**concentrated lithium brine, x% Li**" (*expressed per kg of concentrated brine*).

This scaling is essential because **the concentrated brine is the intermediate product that feeds into lithium carbonate production.**

The scaling approach uses the lithium mass balance between raw brine input and concentrated brine output, along with site-specific lithium concentrations. Using **Salar de Olaroz** as an example, with the data from Schenker & Pfister:

- 1 unit of evaporation pond requires 0.02446 kg of lithium from the environment (raw brine input)
- 1 kg of lithium carbonate battery grade requires 0.348 kg of lithium from the environment (*computing from the cradle-to-gate data reported in Schenker*)
- Therefore, 1 kg of lithium carbonate battery grade requires 14.25 units of evaporation pond (calculated as 0.348 kg of Li per kg Li_2CO_3 / 0.02446 kg of Li per unit)

- The raw lithium brine has a lithium concentration of 0.06% Li by weight according to Schenker & Pfister (2025), Sheet 'B.2_LCI - Input data'
- The concentrated lithium brine has a lithium concentration of 1.20% Li by weight according to Schenker & Pfister (2025), Sheet 'B.2_LCI - Input data'

To calculate **the concentrated brine produced per evaporation pond unit**:

- 29 kg of concentrated brine per kg of lithium carbonate are required (calculated as 0.348 kg of Li per kg Li_2CO_3 / 0.012 kg Li per kg conc. Li brine)
- **2.04 kg of concentrated brine per pond unit** are produced (calculated as 29 kg / 14.25 units of evaporation)

Result: 2.04 kg of concentrated lithium brine (1.2% Li) are produced per 1 unit of evaporation pond at Salar de Olaroz. Therefore, if we use **the reference product "lithium brine, concentrated, 1.2% Li"**, we need to divide the amount of each exchange in the inventory of Schenker and Pfister for the evaporation pond by a factor of 2.04 to express it **per kg of concentrated brine**.

This scaling methodology **was applied consistently across all datasets to ensure that evaporation pond inventories from Schenker & Pfister (2025) are correctly expressed per kg of concentrated brine**, enabling proper integration with downstream lithium carbonate production processes.

Exchanges comments

We plan to improve the exchanges comments inside the ecoinvent datasets.

We had originally structured them to mention from which process step in the model of Schenker the amounts are taken from. However, we are now reconsidering this approach as we think that only mentioning the process step of the model might not be the most straightforward way for users to grasp the role of each exchange. **We plan to work on exchange comments improvement during the first review round, we will then resubmit the datasets with these updates during the second review round together with the addressing of the comments from the reviewer.** We report here **2 examples** of how we plan to work on the comments, from the dataset 'lithium brine concentration, Salar de Olaroz | AR':

1. Current comment to the exchange 'sulfuric acid': From Schenker & Pfister, 2025. SI. Used in the process Water_Ola, df_evaporation_ponds. Scaled to 1 kg of concentrated lithium brine (2.04kg of concentrated lithium brine per kg of evaporation pond)
To implement comment to the exchange 'sulfuric acid': Sulfuric acid is used for pH adjustment during brine processing in evaporation ponds to optimize lithium precipitation and minimize magnesium co-precipitation. From Schenker & Pfister, 2025. SI. Allocated to the process df_evaporation_ponds. Scaled to 1 kg of concentrated lithium brine (2.04 kg of concentrated lithium brine per kg of evaporation pond output).
2. Current comment to the exchange 'quicklime, milled, packed': From Schenker & Pfister, 2025. SI. Used in the process df_evaporation_ponds. Scaled to 1 kg of concentrated lithium brine (2.04kg of concentrated lithium brine per kg of

evaporation pond)

To implement comment to the exchange 'quicklime, milled, packed' ‘:

Quicklime (CaO) is used in the brine concentration stage (evaporation pond system / associated brine treatment) to precipitate and remove dissolved impurities (e.g. Mg) from the brine prior to downstream processing. Amount based on Schenker & Pfister (2025) Supplementary Information, where it is modelled as an input to *df_evaporation_ponds*. Scaled for this dataset to 1 kg concentrated lithium brine, using the factor 2.04 kg concentrated brine per 1 kg evaporation pond output.

Soda ash

This section addresses a general specification issue regarding soda ash.

A discrepancy exists across literature sources regarding whether "soda ash, light" or "soda ash, dense" is used. Schenker & Pfister (2025) explicitly uses "soda ash, light". Kelly et al. (2021) report "soda ash" generically (Table S.6 and S.7, Supplementary Information) without specifying light or dense, but Chordia et al. (2022) explicitly map this to "soda ash, dense".

A soda ash study from the [Argentinian government](#) indicates that natural soda ash is predominantly used for large-volume industrial applications, such as chemical precipitation and mineral processing, and is associated with lower energy use and CO₂ emissions than synthetic soda ash; however, as ecoinvent does not currently provide an explicit dataset for natural soda ash, the background system necessarily relies on synthetic soda ash.

All ecoinvent datasets currently use "**soda ash, dense**". **Action point** We seek the reviewer's advice on the most appropriate choice between "soda ash, light" and "soda ash, dense". This specification has a substantial impact on the IPCC 2021 GWP100 scores (climate change: total, excl. biogenic CO₂), as the two soda ash types have significantly different emission factors in ecoinvent. The influence of this choice on the carbon footprint can be observed in the GWP100 scores presented in the following sections.

Please note that also for the spodumene dataset the exchange 'soda ash, dense' was selected, as also Khakmardan does not specify further the type of soda ash. The effect in spodumene is smaller, as reported in the Scores section of the [spodumene dataset](#).

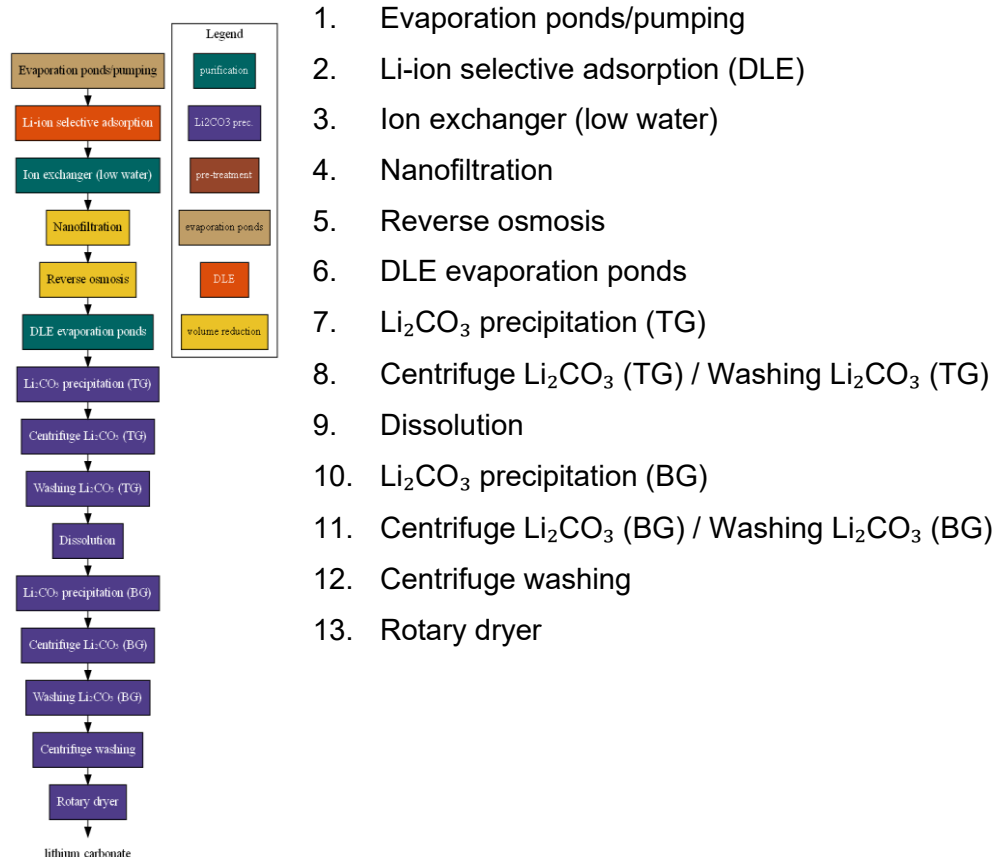
5.2.1 Sites overview

This section presents an overview of the sites covered together with estimation of the GWP100 score of each dataset. Please keep in mind that **these are estimations** based on v3.12 framework and that the final scores will be affected by next iterations in this project and updates in other areas of the database.

5.2.1.1 Chaerhan,CN

Process Overview and Dataset Boundaries

The Chaerhan lithium production route follows a DLE-based process sequence as illustrated below from Schenker & Pfister (2025). The complete production chain includes:



Dataset boundary definition:

- ecoinvent **lithium brine concentration dataset (CN)** covers steps 1-6 in the process sequence (evaporation ponds/pumping through DLE evaporation ponds), producing concentrated lithium brine at 0.022% Li as the reference product. This represents the upstream brine enrichment and concentration processes.
- ecoinvent **lithium carbonate production dataset (CN)** begins at step 7 and covers all downstream processing (Li₂CO₃ precipitation through rotary dryer), using the

concentrated brine (0.022% Li) as input and producing battery-grade Li_2CO_3 as the reference product.

Geography: Chaerhan (Qarhan) salt lake, Qinghai Province, China. Feed brine is reported as brine/pulp from a K-fertilizer plant, with Li concentration 0.25–0.6 g/L.<

Time period: Current operation representative for the early-2020s

Technology:

- DLE-type brine concentration route using Li-ion selective adsorption with adsorption and desorption phases (with heating), followed by downstream volume-reduction/purification through ion exchange and membrane filtration (nanofiltration + reverse osmosis). In the Schenker & Pfister 2025 framework, Chaerhan is highlighted as a special DLE case relying mainly on volume reduction and not reporting a separate pretreatment phase before the DLE column (unlike other DLE sites).

Output: Concentrated lithium brine, 0.022% Li

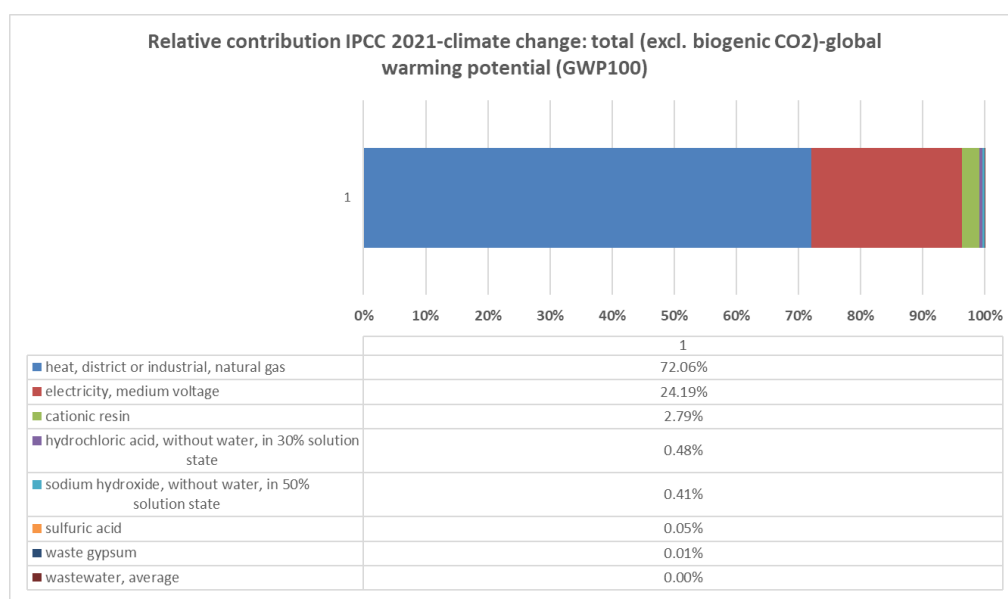
- Battery-grade production is modeled with the brine-to- Li_2CO_3 processing structure shown above from Schenker & Pfister (2025). CO_2 is largely recycled within the upgrading loop as described in the Schenker & Pfister (2025) model documentation. Energy and material calculations follow the Schenker & Pfister (2025) Supplementary Information process modules.

Output: Battery-grade Li_2CO_3

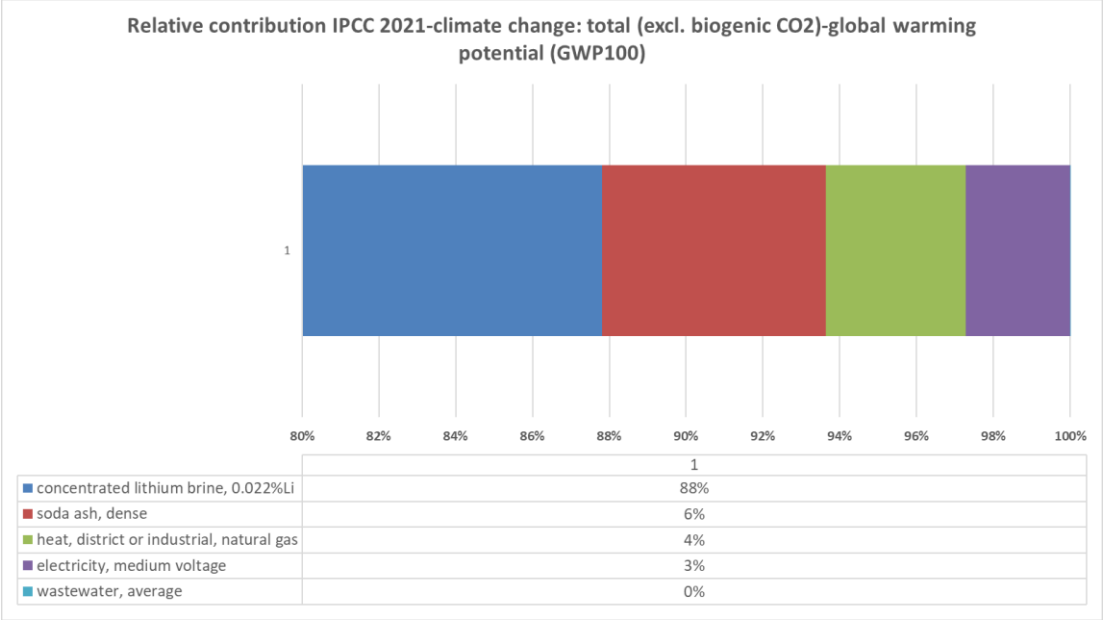
GWP100 Score (IPCC 2021)

No score estimation for the brine is reported in Schenker & Pfister 2025.

ei score estimation 'lithium brine concentration | CN': **0.0314 kg CO₂-Eq**



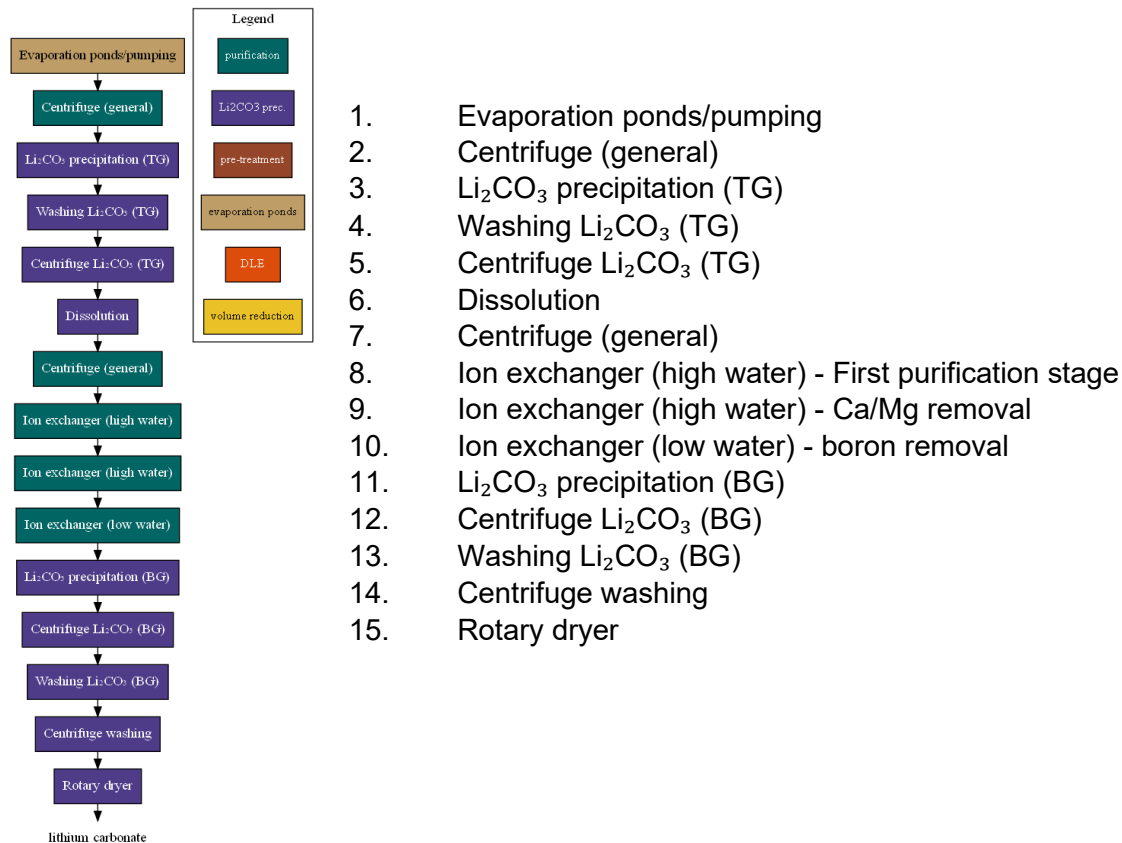
Schenker & Pfister 2025 report for the lithium carbonate battery grade as: **39.1 kg CO2-Eq**
ei score estimation 'lithium carbonate production, battery grade | CN': **39.7 kg CO2-Eq***
**Please refer to the direct contributions lithium.xlsx file for soda ash effect analysis.*



5.2.1.2 Salar de Olaroz, AR

Process Overview and Dataset Boundaries

The Salar de Olaroz lithium production route follows a conventional evaporation pond concentration followed by intensive purification and upgrading sequence, as illustrated in the flowchart below from Schenker & Pfister (2025). The complete production chain includes:



Dataset boundary definition:

- ecoinvent **lithium brine concentration, Salar de Olaroz dataset (AR)** covers step 1 in the process sequence (evaporation ponds/pumping with quicklime addition for Mg/Ca removal), producing concentrated lithium brine at 1.2% Li as the reference product. The process includes volume reduction in evaporation ponds and acidification using sulfuric acid for pH adjustment.
- ecoinvent **lithium carbonate production, battery grade, from concentrated brine, Salar de Olaroz dataset (AR)** begins at step 2 and covers all downstream processing (Li_2CO_3 precipitation through rotary dryer), using the concentrated brine (1.2% Li) as input and producing battery-grade Li_2CO_3 as the reference product. This stage includes the intensive ion exchanger treatment (steps 8-10) that characterizes Olaroz operations.

Geography: Salar de Olaroz, Argentina

Time period: Dataset represents operations from 2011-2019, with time validity set to 2019 to reflect the key operational reporting year used for Olaroz assumptions.

Technology:

- Conventional salar concentration via evaporation ponds. Brine is pumped from 50 m below ground and concentrated by solar evaporation to ~1.2 wt% Li. The process includes volume reduction in evaporation ponds with quicklime addition for magnesium and calcium removal, followed by acidification using sulfuric acid for pH adjustment and centrifugation for solid-liquid separation.
Output: Concentrated lithium brine, 1.2% Li
- Conventional brine-route upgrading to battery grade via two-stage precipitation/upgrading. Technical-grade Li_2CO_3 is precipitated with soda ash (modeled stoichiometrically with surplus to account for losses in Schenker & Pfister), followed by washing and centrifugation. The technical-grade Li_2CO_3 is then dissolved at low temperature with CO_2 to improve dissolution. Before battery-grade precipitation, the solution undergoes intensive purification through ion exchangers. The intensive purification through three ion exchangers during the carbonate production stage is an Olaroz-specific process characteristic: two for Ca/Mg removal (high water) and one for boron removal (low water). Battery-grade Li_2CO_3 is then precipitated with CO_2 recirculation, followed by centrifugation, washing, and drying (rotary dryer).
Output: Battery-grade Li_2CO_3

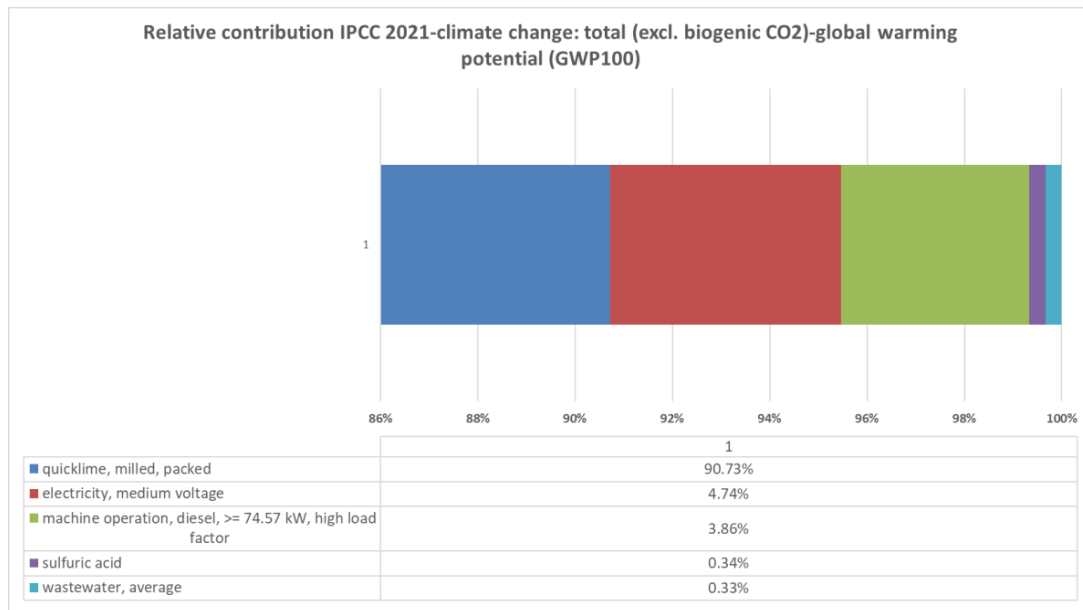
Complementary information:

From direct communication with the authors, we know that the model for Olaroz is very likely underestimating salt tailings (which are reported at 9.9e-17 kg of salt tailings at the evaporation pond). This is due to lack of reported enriched brine chemistry in the evaporation ponds at the site. As a result, the reported values for salt tailings in the publication are too and should not be considered confirmed.ecoinvent dataset does not include salt tailings.

GWP100 Score (IPCC 2021)

No score estimation for the brine is reported in Schenker & Pfister 2025.

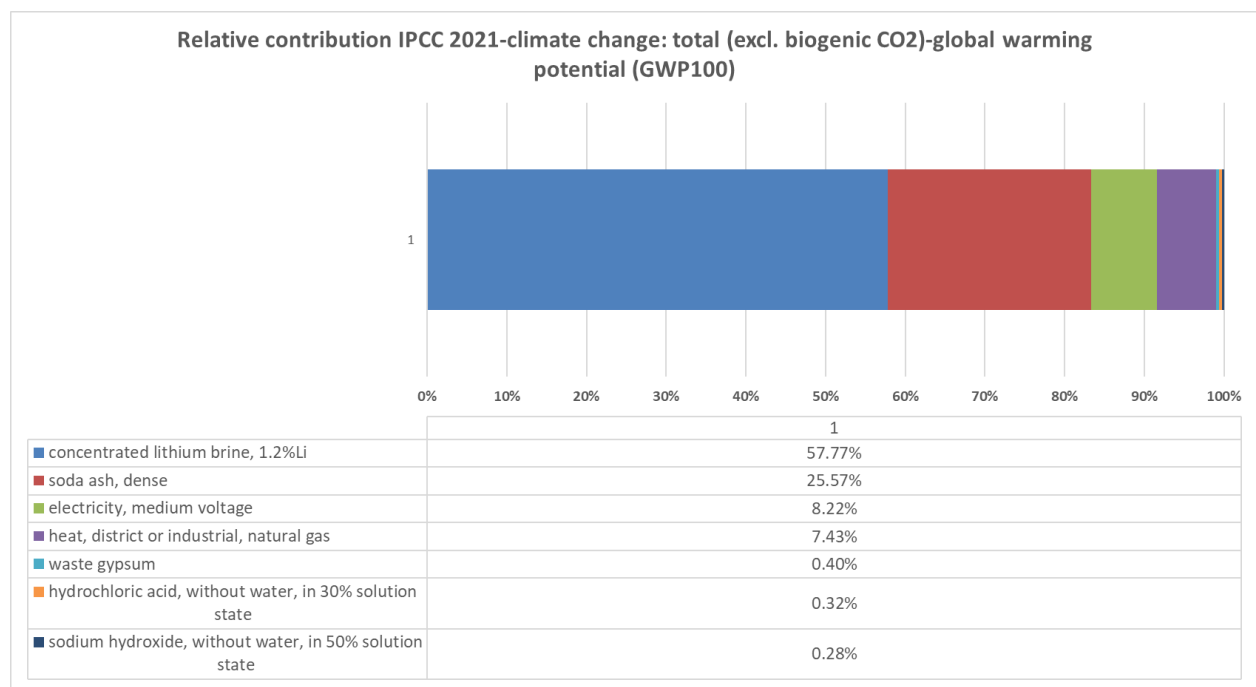
ei score estimation 'lithium brine concentration, Salar de Olaroz | AR: **0.179 kg CO2-Eq**



Schenker & Pfister 2025 report for the lithium carbonate battery grade as: **7.2 kg CO2-Eq**

ei score estimation 'lithium carbonate production, battery grade, from concentrated brine, Salar de Olaroz | AR: **9.044 kg CO2-Eq**

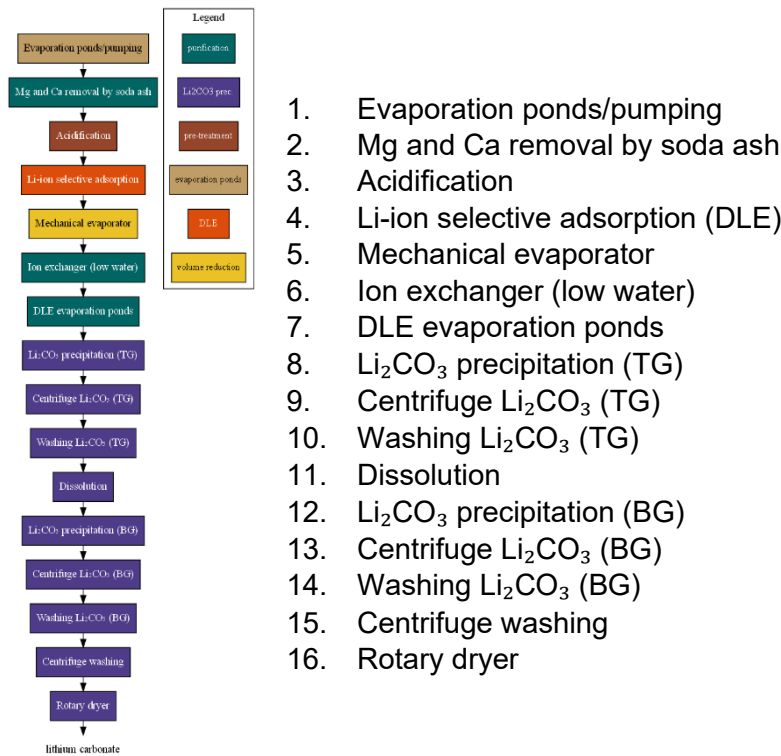
**Please refer to the direct contributions lithium.xlsx file for soda ash effect analysis.*



5.2.1.3 Fenix, AR

Process Overview and Dataset Boundaries

The Fenix lithium production route follows a DLE-based concentration with evaporation pond pre-treatment as illustrated in the flowchart below from Schenker & Pfister (2025). The complete production chain includes:



Dataset boundary definition:

- ecoinvent **lithium brine concentration, Fenix dataset (AR)** covers steps 1-7 in the process sequence (evaporation ponds/pumping through ion exchanger for low-water purification), producing concentrated lithium brine at 0.7% Li as the reference product. This represents the upstream brine enrichment through DLE extraction and purification.
- ecoinvent **lithium carbonate production, battery grade, from concentrated brine, Fenix dataset (AR)** begins at step 8 and covers all downstream processing (Li₂CO₃ precipitation through rotary dryer), using the concentrated brine (0.7% Li) as input and producing battery-grade Li₂CO₃ as the reference product.

Geography: Fenix brine operation at Salar del Hombre Muerto, Argentina. Direct lithium extraction (ion-exchanger/resin adsorption) route as described in Schenker et al. (2022) SI as an example of site-specific technology used by Livent.

Time period: Current operations around 2022 as modeled in Schenker & Pfister (2025)

Technology characterization:

- Fenix follows a DLE-based concentration route that combines conventional evaporation ponds with advanced DLE technology. The process sequence includes initial solar evaporation for volume reduction, followed by chemical pre-treatment (soda ash for Mg/Ca removal), DLE Li-selective adsorption for efficient lithium extraction, mechanical evaporation for further concentration, and ion exchange for final purification. This hybrid approach distinguishes Fenix from purely conventional evaporation pond operations (like Olaroz and Silver Peak) and from DLE operations without evaporation pond pre-treatment (like Chaerhan).

Output: Concentrated lithium brine, 0.07%Li

- Brine-route production of battery-grade Li_2CO_3 following the Schenker & Pfister (2025) modeling framework. CO_2 recirculation is modeled as described in the Supplementary Information of Schenker & Pfister; heat losses are represented as waste heat where applicable.

Output: Battery-grade Li_2CO_3

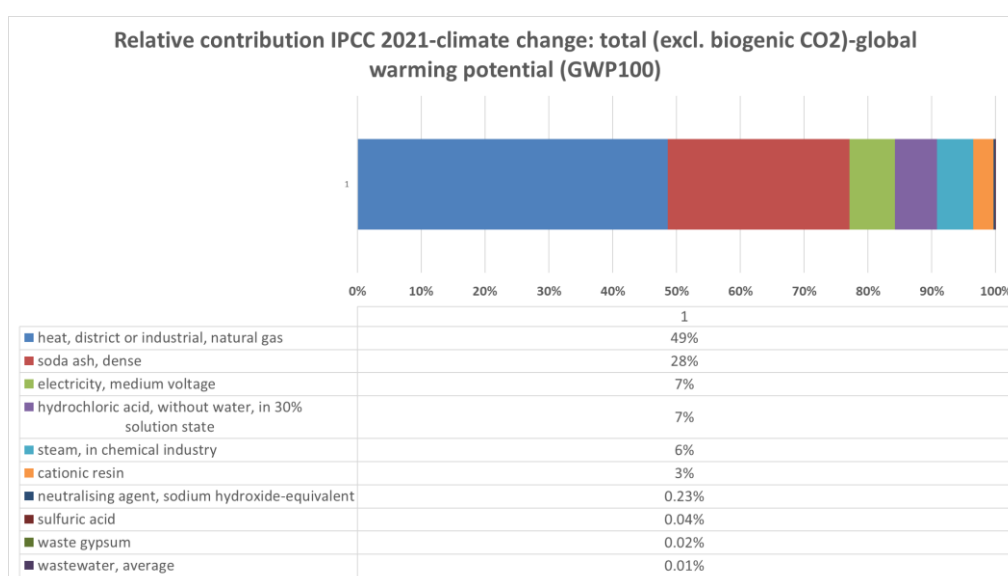
Complementary notes:

DLE evaporation ponds data gap: From direct communication with the authors, the DLE evaporation pond component is not fully inventoried in the current model for Fenix and other DLE sites. While these facilities employ DLE technologies with evaporation ponds in the processing scheme, insufficient data exist to quantify pond sizing, salt crystallization processes, and associated land occupation and transformation. As a result, the Fenix inventory in the 2025 SI lacks occupation, transformation, and salt tailing exchanges. The authors note that recent technical documentation from Arcadium/Livent may enable future estimation of these evaporation pond parameters, representing a potential improvement area for future model iterations.

GWP100 Score (IPCC 2021):

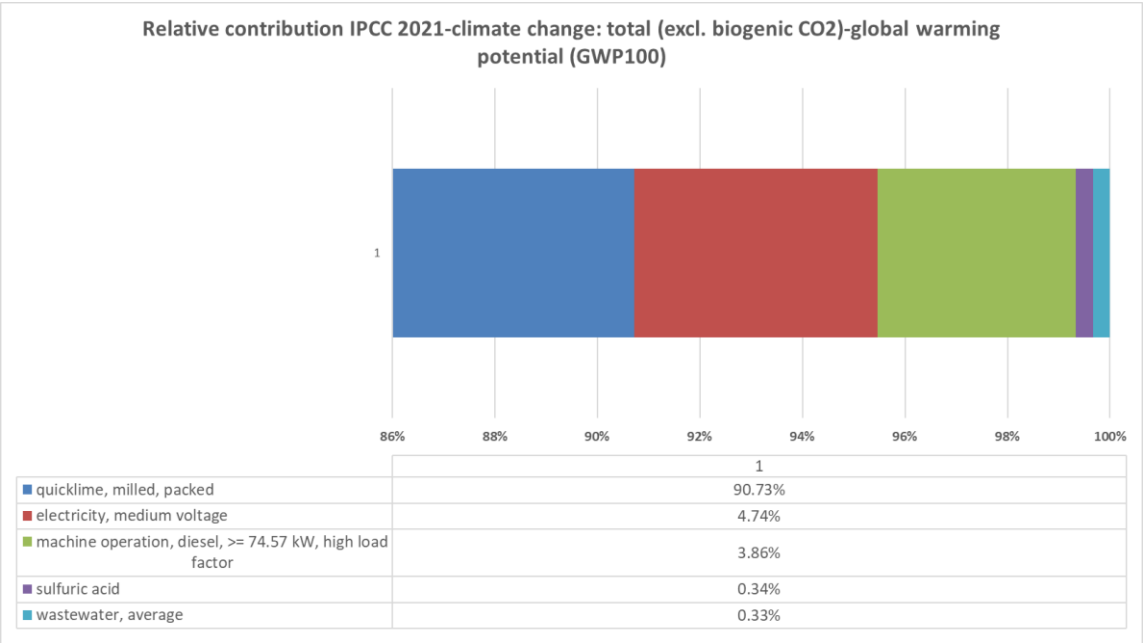
No score estimation for the brine is reported in Schenker & Pfister 2025.

ei score estimation 'lithium brine concentration, Fenix | AR: **0.0397 kg CO₂-Eq**



Schenker & Pfister 2025 report for the lithium carbonate battery grade as: **29.6 kg CO2-Eq**
ecoinvent score estimation 'lithium carbonate production, battery grade, from concentrated
brine, Fenix | AR: **36.5 kg CO2-Eq**

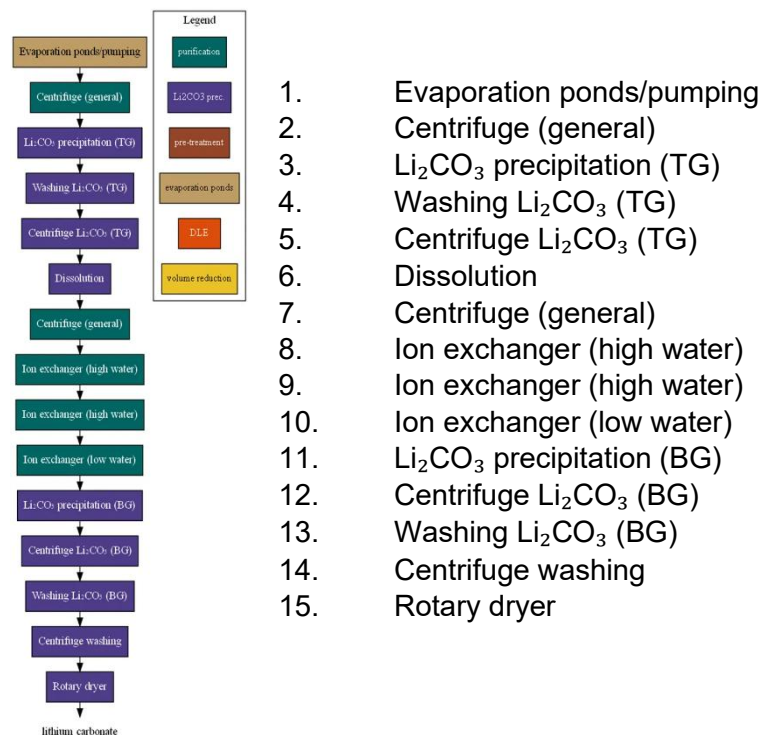
**Please refer to the direct contributions lithium.xlsx file for soda ash effect analysis.*



5.2.1.4 Silver Peak

Process Overview and Dataset Boundaries

The Silver Peak lithium production route follows a conventional evaporation pond concentration with ion exchanger upgrading, as illustrated in the flowchart below from Schenker & Pfister (2025). The complete production chain includes:



Dataset boundary definition:

- ecoinvent **lithium brine concentration dataset (US)** covers step 1 in the process sequence (evaporation ponds/pumping with quicklime addition for impurity removal), producing concentrated lithium brine at 0.6% Li as the reference product. Raw brine is pumped to surface and concentrated in a pond cascade through solar evaporation.
- ecoinvent **lithium carbonate production, battery grade, from concentrated brine dataset (US)** begins at step 2 and covers all downstream processing (centrifuge through rotary dryer), using the concentrated brine (0.6% Li) as input and producing battery-grade Li_2CO_3 as the reference product. This stage includes the three ion exchanger purification steps (steps 8-10) that characterize the Silver Peak ion-exchanger route.

Geography: Silver Peak (Clayton Valley), Nevada, United States

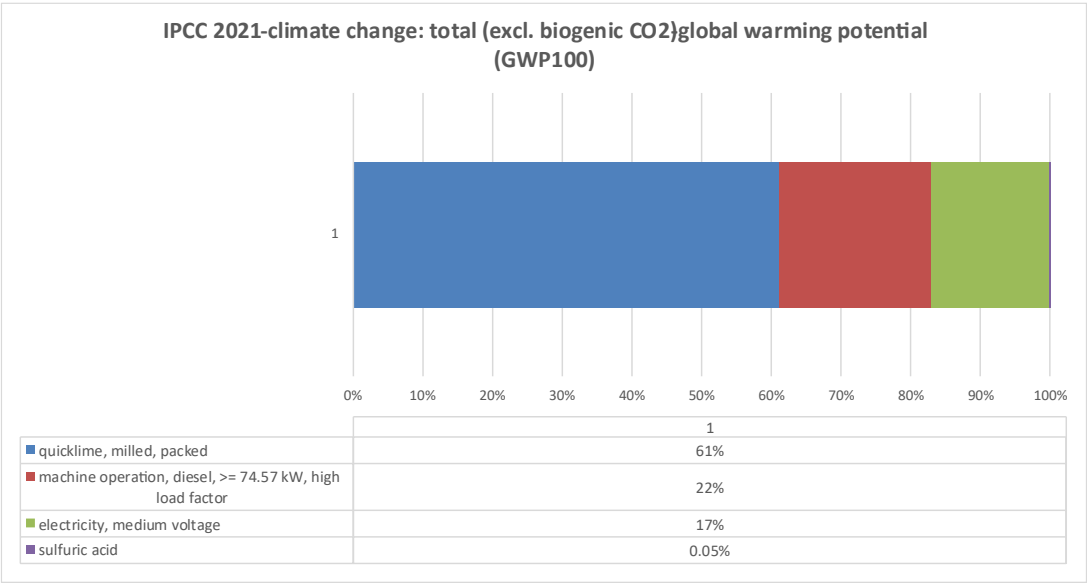
Time period: Current operations as compiled in Schenker & Pfister (2025), dates set to 2022

Technology characterization:

- Silver Peak follows a conventional solar concentration route via evaporation ponds with solar energy as the primary driver for water evaporation. Raw brine is pumped to surface and concentrated in a pond cascade to 0.6% Li, with quicklime used for impurity removal during the evaporation stage.
Output: lithium brine 0.6%
- The downstream processing is characterized by a conventional ion-exchanger route with three ion exchange purification stages before battery-grade precipitation.
Output: Battery-grade Li_2CO_3

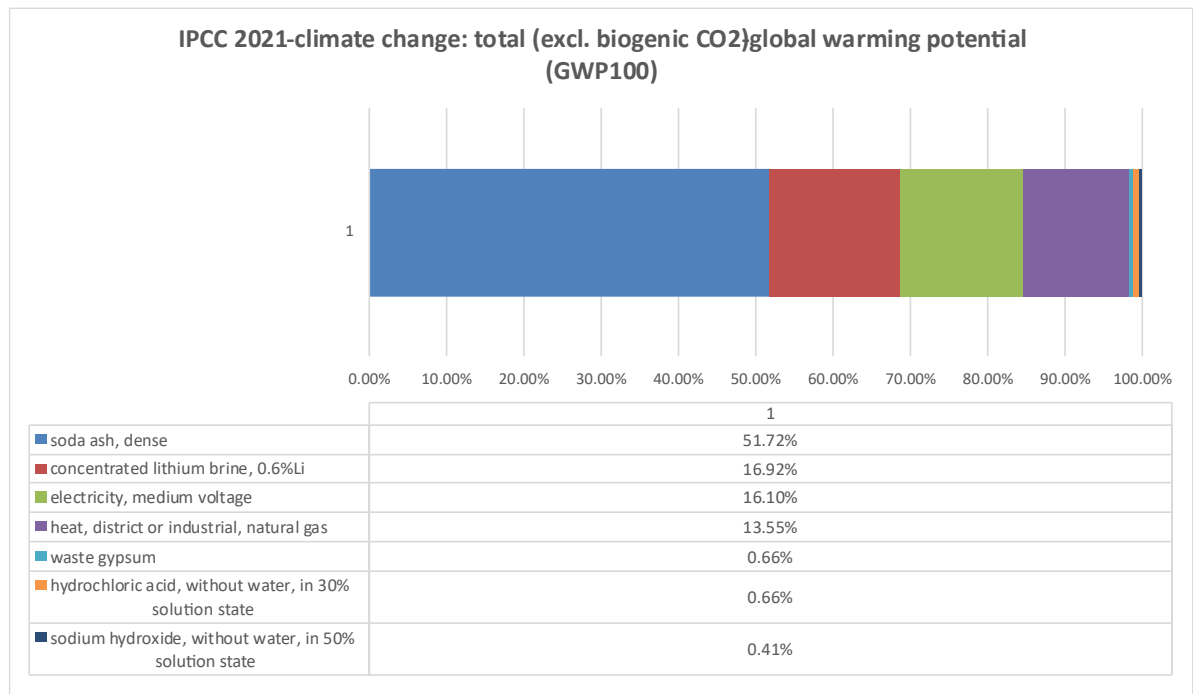
GWP100 Score (IPCC 2021):

No score estimation for the brine is reported in Schenker & Pfister 2025.
ei score estimation 'lithium brine concentration, | US: **0.051 kg CO2-Eq**



Schenker & Pfister 2025 report for the lithium carbonate battery grade as: **2.8 kg CO2-Eq**ecoinvent score estimation 'lithium carbonate production, battery grade, from concentrated brine | US : **4.47 kg CO2-Eq** *

**Please refer to the direct contributions lithium.xlsx file for soda ash effect analysis.*



5.2.2 Mass balance in lithium brine concentration datasets

A document in support of the mass balance analysis has been shared with this report **unit process balance (brine concentration, undef).xlsx**.

In the Excel document, it is possible to observe for each property whether there is a balance between the input and output, and which exchanges mostly contribute to it. The Excel shows 1 sheet for each site.

An important methodological consideration affects the interpretation of these mass balances. The source literature (Schenker & Pfister, 2025) follows the AWARE methodology for water scarcity assessment, as recommended by the UNEP-SETAC Life Cycle Initiative. This methodology specifically **excludes saline brine water** from water consumption accounting, focusing only on freshwater consumption.

As stated in Schenker & Pfister (2025, p. E): "*Kelly et al., Chordia et al., and Schenker et al. do not include the water content of the brine in their assessments when assessing water scarcity. As AWARE is designed to only assess freshwater consumption, we follow the approach defined by the UNEP-SETAC Life Cycle Initiative.*"

A complete mass balance for lithium brine concentration should account for:

Inputs:

- Raw lithium brine (with specified water content and dissolved mineral composition)
- Freshwater (for washing, dust suppression, or dilution)
- Process chemicals (acids, bases, flocculants as applicable)

Outputs:

- Concentrated lithium brine (with specified water content and mineral composition)
- Evaporated water
- Precipitated salt tailings (primarily NaCl, but also K, Mg, Ca, and sulfate salts)

Current data gaps affecting mass balance

Beyond the exclusion of brine water from water consumption accounting, additional data gaps contribute to the incomplete mass balance in these datasets:

Missing mineral extraction flows: Because the brine water itself is not reported as an input flow (consistent with AWARE), the dissolved minerals carried in that brine water are also not explicitly reported as extractions from ground.

Incomplete salt tailings reporting: As communicated by the dataset authors, salt tailings data is incomplete or missing in the current datasets.

Planned actions on the ecoinvent side

To address the identified mass balance issues while respecting the methodological choices in the source literature, we plan to implement the following improvements:

1. Include lithium brine composition in the reference product

At the moment, we included in the Reference Product properties only the lithium content.

We will incorporate the brine composition data (water content and dissolved mineral concentrations) as properties of the concentrated lithium brine reference product, based on the data available in Schenker & Pfister (2025), Sheet '**B.2_LCI - Input data**'.

This will provide users with transparent information about **the actual composition of the output brine**. In particular, the water content will be accounted for.

Action point: Brine composition data is not available for all sites in the Schenker dataset. For sites where composition data is missing (Olaroz, Silver Peak and Fenix), **we seek editor advice on selecting an appropriate proxy for brine composition when missing**. The sheet '**B.2_LCI - Input data**' has been reported in the attached Excel **unit process balance (brine concentration, undef).xlsx**.

2. Document methodology and data gaps

We propose to clearly document in the dataset metadata:

- The application of the AWARE methodology for water accounting, which deliberately excludes saline brine water from water consumption metrics
- The current data gaps regarding precipitated salts and complete water balance

This transparent documentation will help LCA practitioners understand the limitations of these datasets.

3. Parallel improvements for current or future releases

In parallel with the above actions, we will continue working on the following aspects, either during this release or in subsequent updates:

- **New biosphere flows:** Consider introducing a new elementary flow such as "water, brine, from ground" to distinguish ancient saline brine water from freshwater extraction, allowing for complete water accounting **while maintaining compatibility with AWARE methodology**
- **Salt tailings data:** Work to fill the data gaps on precipitated salts, potentially by reaching out to the dataset authors (Schenker & Pfister) for clarification on additional calculations or proxy data
- **Complete mineral balance:** Develop a more comprehensive approach to tracking all dissolved minerals extracted with the brine and their fate during the concentration process

These parallel improvements would enhance the datasets' utility for a broader range of LCA applications while maintaining their validity for water scarcity assessment under AWARE guidelines.

5.2.3 Mass balance in lithium carbonate, battery grade datasets

A document in support of the mass balance analysis has been shared with this report: unit process balance (Li₂CO₃, undef).xlsx.

The mass balance framework described in [section 5.2.2](#) for lithium brine concentration datasets applies similarly to lithium carbonate production datasets. The same methodological considerations regarding AWARE methodology, water accounting, and data gaps are relevant here.